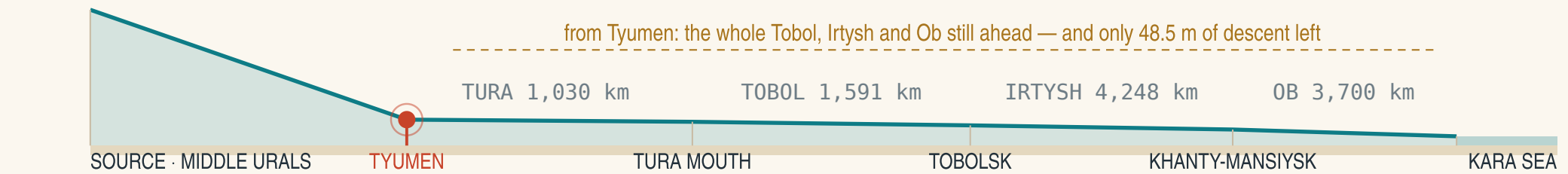


THE HIGH BANK

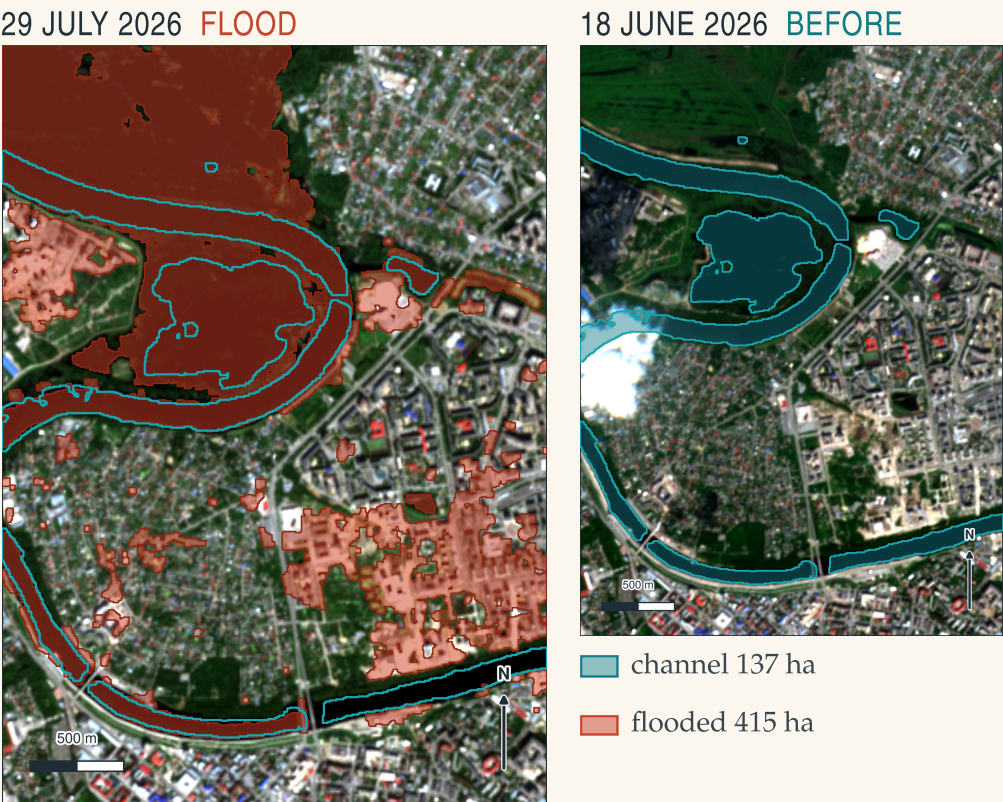
Tyumen was founded in 1586 on the steep southern bank of the Tura and left the low northern bank alone. Every old town chose a side of its river. **Four hundred and forty years on, does that choice still decide who floods?**

2,000 KILOMETRES TO FALL 48 METRES

vertical scale true · horizontal schematic



TWO DAYS BEFORE THE PEAK — *what Sentinel-2 saw over 2.9 km of the Tura*



Orange: water on 29 July. Teal: the normal channel. On the north bank the flood is across the streets.

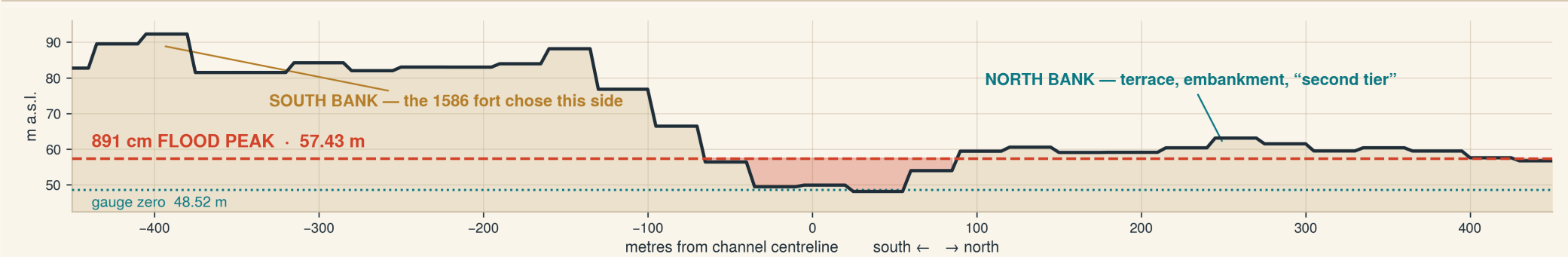
3×
the river's footprint at the peak. The gauge read **891 cm** on 31 July — its highest ever recorded.

Water was mapped with MNDWI and then constrained by terrain, because cloud shadow suppressed the index on the flood date: a pixel counts as flood only where the ground also lies below the observed flood level and connects to the channel.

The 29 July image is the last cloud-free pass before the peak — every number on this poster is a **lower bound**.

NOW TURN THE RIVER NINETY DEGREES

900 m across the valley · Copernicus DEM GLO-30



Two banks, four hundred metres apart, **FORTY TIMES THE RISK**

NORTH BANK <i>low terrace · the side nobody fortified</i>	SOUTH BANK <i>the 1586 bluff · historic centre</i>
16.8% <i>of its buildings flooded (120 of 716)</i>	0.4% <i>of its buildings flooded (18 of 4,437)</i>
391 ha flooded	24 ha flooded
~5,080 people	~560 people

A THRESHOLD, NOT A BOUNDARY

The south bank holds **six times more buildings** — almost none went under. The sixteenth-century choice still sorts the risk. But the north-bank “**second tier**”, built above the floodplain and never flooded in living memory, went under too. The 1586 logic is not a wall — it is a **gradient**, and 2026 found its height.

THIS IS NOT ONLY A SIBERIAN PROBLEM

English towns were sited on the same reasoning. The risk is rarely in the old core — it sits in what was built later, on ground earlier generations left alone.

6.3 million

English properties are in areas at risk of flooding — potentially **8 million, one in four**, by mid-century.

TEWKESBURY, SEVERN & AVON

Its settlers used the “natural rises in the land” to avoid flooding. Abbey and old town still stand on that raised ground — the modern town around them flooded in 2007.

SHREWSBURY, SEVERN

The medieval core sits inside a meander loop chosen for defence — the loop that now holds water against the town. Frankwell flooded three times in six weeks in 2000.